# AMAN KUMAR

Software Engineer | BCA Graduate
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## Professional Summary

BCA graduate (CGPA 8.4) with hands-on experience building and deploying full-stack web applications using React, Node.js, Express, and MongoDB, along with Python/Flask services, RESTful APIs, and JWT authentication. Deployed projects live on Render and Vercel. Strong foundation in Data Structures and Algorithms, OOP, and DBMS; placed 3rd in a coding and debugging competition. Seeking an entry-level Software Engineer role.

## Technical Skills

**Languages:** Python, JavaScript, SQL
**Frontend:** React
**Backend:** Node.js, Express.js, Flask, FastAPI, SQLAlchemy
**Databases:** MongoDB, MySQL, PostgreSQL, SQLite
**Cloud & Deployment:** Docker, Render, Vercel, MongoDB Atlas, Neon
**Developer Tools:** Git, GitHub (Version Control), REST API Development, JWT, Unit Testing (pytest)
**Core CS:** Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS
**Soft Skills:** Team Collaboration, Communication

## Projects

**Job Application Tracker** (GitHub – Live Demo) Oct 2026
*React, Node.js, Express, MongoDB (Schema Design, Indexing), JWT, bcrypt, REST APIs, Render, Vercel*
- Developed a full-stack job application tracker with a React frontend and a Node.js/Express backend exposing 9 REST API endpoints (authentication and CRUD) on a MongoDB schema indexed by user.
- Implemented JWT authentication (7-day token expiry) with bcrypt password hashing and auth middleware, so each user can access only their own job data.
- Added status filtering, company/role search, and a MongoDB aggregation endpoint powering a status pipeline dashboard (Applied, Interview, Offer, Rejected).
- Deployed on MongoDB Atlas (database), Render (backend), and Vercel (frontend); verified the full flow end-to-end on the live URLs.

**Mini Search Engine** (GitHub – Live Demo) Oct 2026
*Python, Flask, SQLite (DBMS), Data Structures, Inverted Index, TF-IDF, Render*
- Engineered a search engine in Python using an inverted index data structure with smoothed TF-IDF ranking, plus a tokenizer that lowercases text and removes stopwords.
- Applied DBMS concepts to design a 2-table SQLite schema (documents, inverted_index) with an index on the word column for fast term lookups.
- Built 4 Flask API endpoints (search, list, add, delete documents) and a web UI; newly added documents are indexed and searchable immediately.
- Seeded 10 documents; ranked queries return in under 1 ms locally; deployed live on Render and confirmed add-document and search in production.

**URL Shortener API** (GitHub – Live Demo) Oct 2026
*Python, FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Docker, pytest, Render*
- Built a REST API with FastAPI that shortens URLs, redirects short codes (HTTP 307) to the original link, and tracks click statistics, backed by PostgreSQL through the SQLAlchemy ORM.
- Designed a 2-table schema (urls, clicks) with a unique index on the short code and an indexed foreign key; click counts use atomic SQL increments to avoid lost updates under concurrent requests.
- Generated random 6-character base62 codes with collision retry and added validated custom aliases (Pydantic URL validation, 409 on duplicates, reserved-word protection).
- Wrote 10 pytest tests on an isolated in-memory database; containerized with Docker and deployed on Render with a Neon PostgreSQL database.

## Certifications

- Problem Solving (Intermediate) – HackerRank (2026)
- Python with DSA – Unstop (2026)
- GitHub Workflow Mastery (2026)

## Achievements & Extracurriculars

- Secured 3rd Rank in Coding and Debugging Competition as part of a 3-member team, IT Fest 2024, Acharya Institute of Graduate Studies, Bengaluru
- Volunteered at IT Quiz, 2024, Acharya Institute of Graduate Studies, supporting event coordination
- Participated in NSS “Walk to EmpowHER”, a women empowerment initiative, 2024

## Education

**Acharya Institute of Graduate Studies, Bengaluru** **2023 – 2026**
Bachelor of Computer Applications (BCA)
CGPA: 8.4 / 10